

SOEN 203 – Introduction to Database Systems

Kisii University | Clean Revision Notes

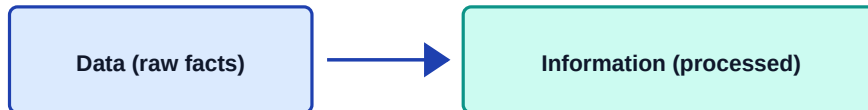
1. Data, Information and Database

Data

Raw facts about something. Data by itself has little meaning until it is organised or processed. Example: the numbers 85, 90, 78 or the name "Jane".

Information

Processed data that has meaning and is useful for decision-making. Example: "Jane scored an average of 84.3% in the CAT."



Data becomes Information when it is processed and given context

Database

A collection of logically related data that is designed to meet the information needs of an organisation. It is an organised store of processed data kept in one place so that it can be easily accessed, managed and updated.

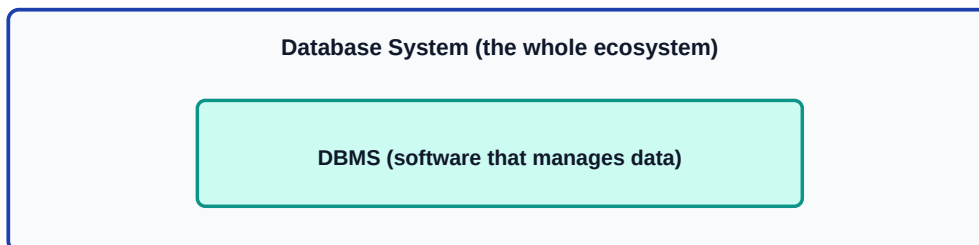
2. DBMS and Database System

Database Management System (DBMS)

Software that is used to create, define, manipulate and manage databases. It acts as the interface between the users/application programs and the actual data stored on disk. Examples: MySQL, PostgreSQL, Oracle, SQL Server, MongoDB.

Database System

The entire ecosystem that works together: the DBMS software, the actual data, the hardware, the people who use and manage it, and the procedures/rules that govern its operation.



DBMS is the software core; the Database System includes everything around it

3. Components of a Database System

A complete database system has five main components that must work together:



The five components that work together in a Database System

- **Hardware** – computers, storage devices, network equipment
- **Software** – the DBMS itself plus operating system and application programs
- **Data** – the most important part; the actual facts stored in the database

- **Procedures** – rules and instructions that govern how the system is used and operated
- **People** – everyone who designs, manages or uses the database

People who deal with databases

- **Database Administrator (DBA)** – manages the database, controls access, ensures performance and security
- **Database Designer** – designs the structure of the database (tables, relationships, etc.)
- **End User** – people who use the data for their daily work (query, update, generate reports)
- **System Analyst** – studies requirements and helps decide how the database should support the organisation
- **Application Programmer** – writes programs that interact with the database

4. Common Types of Databases

Relational Database (most common)

Stores data in related tables. Each table has rows (records) and columns (attributes). Tables are linked using unique keys (primary keys and foreign keys). Uses Structured Query Language (SQL), which was developed around 1970. Examples: MySQL, PostgreSQL, Oracle, SQL Server.

Unstructured / NoSQL Database

Suitable for data that does not fit neatly into tables — for example text messages, social media comments, documents, sensor data. Good for large volumes of rapidly changing or irregular data. Example: MongoDB.

Cloud Database

Database that is hosted on remote servers and accessed over the Internet. The provider manages hardware, backups and scaling. Examples: Amazon RDS / Aurora, Google Cloud SQL, Azure SQL Database.

5. Main Functions of a Database / DBMS

- **Data storage, retrieval and update** – the core purpose: store data safely and get it back when needed
- **User-accessible catalogue** – a system catalogue (data dictionary) that describes the structure of the data
- **Authentication & security services** – control who can log in and what they are allowed to do
- **Integrity services** – enforce rules so that data remains accurate and consistent (e.g. a mark cannot be 150%)
- **Concurrency control services** – allow many users to work at the same time without interfering with each other
- **Utility services** – backup, recovery, import/export, monitoring, etc.
- **Audit / transaction log** – keeps a record of who changed what, when and from where (useful for security and recovery)

6. Advantages of Using a Database

- **Controlled data redundancy** – data is stored once (or with minimal duplication) instead of being repeated in many files
- **Data consistency** – because data is not duplicated everywhere, updates are made in one place and stay consistent
- **Sharing of data** – many users and applications can access the same data at the same time
- **Improved data integrity** – rules (constraints) can be enforced by the DBMS so that only valid data is stored
- **Concurrency** – many people can use the database simultaneously without corrupting the data
- **Consistency across platforms** – data can be accessed from different applications and devices in a controlled way
- **Security** – access can be restricted to authorised users only
- **Backup and recovery** – modern DBMS provide tools to protect data against loss

7. Disadvantages / Challenges

- **Financial cost** – software licences, hardware, skilled staff and ongoing maintenance can be expensive
- **Complexity and maintenance** – designing, installing and looking after a large database system requires expertise
- **Vulnerabilities** – if security is weak, a central database can become a single point of attack or failure
- **Performance overhead** – the extra layers of the DBMS and network traffic can sometimes make simple operations slower than a small local file

8. Quick Revision Checklist

- Data = raw facts; Information = processed, meaningful data.
- Database = organised collection of related data.
- DBMS = software that creates and manages databases.
- Database System = DBMS + data + hardware + people + procedures.

- Five components: Hardware, Software, Data, Procedures, People.
- Relational databases use tables and SQL; NoSQL handles unstructured data; Cloud databases are hosted remotely.
- Key advantages: less redundancy, consistency, sharing, integrity, concurrency, security.
- Key disadvantages: cost, complexity, security risks, possible performance overhead.

These notes expand the class material for SOEN 203. Use the diagrams to remember the difference between data and information, the five components, and how DBMS fits inside the larger Database System.